

CSE 4345: Big Data Analytics

Week 1, Part 2 — Seminal Research & Case Study

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Today's Agenda

Seminal Papers Covered

- 1 Laney, D. (2001). *3D Data Management*. META Group.
- 2 Brewer, E. (2000). *Towards Robust Distributed Systems*. PODC.
- 3 Gilbert & Lynch (2002). *Brewer's Conjecture*. ACM SIGACT.

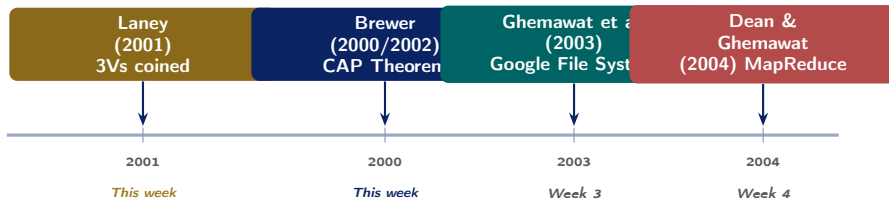
Case Study

Manyika et al. (2011). *Big Data: The Next Frontier for Innovation, Competition, and Productivity*. McKinsey Global Institute.

CLO Addressed

Research Paper 1 — Laney (2001): The 3Vs Origin

Research Landscape: Why These Papers?



Reading Sequence

Laney (2001) defines *what* Big Data is → Brewer/Gilbert (2000–2002) defines *why* distributed design is hard → GFS (2003) and MapReduce (2004) provide the *engineering solutions*. This course follows that exact intellectual lineage.

Laney (2001): Publication Context

Full Citation

Laney, D. (2001, February 6). **3D Data Management: Controlling Data Volume, Velocity and Variety**. META Group Application Delivery Strategies Research Note, File 949.

Venue: META Group analyst memo (2 pages)

Cited by: >20,000 times (Google Scholar, 2024)

Impact: Became the canonical industry and academic definition of Big Data

Laney's Own Reflection (2012 Interview)

"It was sort of a throwaway piece. I had been thinking about the data challenges at e-commerce companies and wrote down three dimensions. I never expected it to become the definition of an entire era of computing."

—Doug Laney, Gartner Blog, 2012

The Author

Doug Laney was a research analyst at META Group (later acquired by Gartner in 2005). He wrote this as a short practitioner note about **e-commerce data management challenges**—not a formal academic

Why Study a 2-page Memo?

It illustrates that **foundational frameworks** need not come from prestigious venues. The quality of the *insight* matters more than the journal. This is a recurring theme in Big Data research.

Laney (2001): The Original 3Vs—Close Reading

Opening Argument (Paraphrased)

“E-commerce has exploded the volume, velocity and variety of information that companies must manage. Retailers and their trading partners face the challenge of managing large, fast-moving and diverse data from the Web, RFID, point-of-sale sensors and order entry systems.”

Laney’s original definitions:

- **Volume:** Transactions at an online retailer generate **exponentially more** data than equivalent brick-and-mortar operations. Click-stream data, session logs, product views—each visit generates thousands of records.
- **Velocity:** The **time-sensitivity** of data. Stock prices, fraud signals, and real time offers must be

The Insight That Made This Seminal

Laney was the first to observe that these three dimensions are **interdependent and compound**:

- High velocity + high variety = integration is nearly impossible in real time
- High volume + high variety = storage schemas must be flexible
- High volume + high velocity = single-node processing cannot keep up

This **3D problem space** is why no single traditional tool solves Big Data—you need an *ecosystem*.

Laney (2001): The E-Commerce Motivation

The year is 2001. The dot-com boom has just ended. What data problems were practitioners actually facing?

Laney's Observed Challenges

Dimension	Observed Problem (2001)
Volume	Amazon.com: millions of product pages, billions of click-stream events per month. No RDBMS could store it all.
Velocity	eBay: real-time auction bids must update in seconds. Batch-nightly processing was useless.

2024 Parallel: Same Problem, 1,000× Scale

- **Volume (2024):** Amazon: >1.6 million orders per day; petabytes of product images, reviews, fulfillment data
- **Velocity:** Daraz.com.bd: flash-sale order spikes of 10,000 orders/minute require real-time inventory sync
- **Variety:** Shopper GPS + purchase history + product image + chatbot transcript + payment gateway log = five different formats in one customer journey

The Enduring Relevance

Laney's 3Vs from 2001 describe a problem

Laney (2001): Legacy, Extensions, and Critique

What the paper triggered:

- Gartner adopted 3Vs as official Big Data definition (2012)
- IBM extended to **4Vs** (adding Veracity, 2013)
- IDC added **Value** as the 5th V (2014)
- Over 40 papers have proposed additional Vs (Validity, Volatility, Venue, Vocabulary...)

The 5Vs Consensus (Gartner / IDC, 2012–2014)

V	Added by
Volume, Velocity, Variety	Laney (2001)
Veracity (quality)	IBM (2013)
Value (utility)	IDC (2014)

Academic Critique of the 3Vs

The 3Vs are a **useful taxonomy**, not a formal theory:

- They describe *symptoms*, not solutions
- They do not specify *thresholds*: how much volume is “big”?
- They ignore *social and ethical* dimensions of data (privacy, consent, bias)
- Kitchin (2014) argues they reinforce a **techno-deterministic** view that more data always produces better insights

Reference: Kitchin, R. (2014). *The Data Revolution*. Sage. Ch. 4.

Research Paper 2 — Brewer (2000) & Gilbert & Lynch (2002): CAP Theorem

Brewer (2000): The CAP Conjecture

Full Citations

- [1] Brewer, E. A. (2000). **Towards Robust Distributed Systems**. Invited Keynote, ACM Symposium on Principles of Distributed Computing (PODC), Portland, Oregon.
- [2] Gilbert, S. & Lynch, N. (2002). **Brewer's Conjecture and the Feasibility of Consistent, Available, Partition-Tolerant Web Services**. ACM SIGACT News, 33(2), 51–59.

About the Authors

Eric Brewer: UC Berkeley professor; co-founder of Inktomi (early web search engine), VP of Infrastructure at Google. His *practitioner* instincts drove the conjecture.

The Conjecture (Brewer, 2000)

"It is impossible for a web service to simultaneously provide all three of the following guarantees:

- **Consistency:** *all nodes see the same data at the same time;*
- **Availability:** *a guarantee that every request receives a response;*
- **Partition tolerance:** *the system continues to operate despite arbitrary message loss or failure of part of the system."*

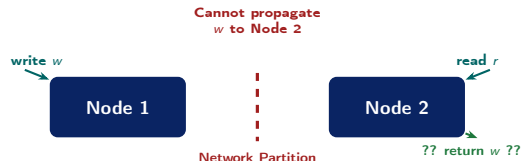
— Brewer, PODC 2000 Keynote

Gilbert & Lynch (2002): The Formal Proof

Proof Strategy: Contradiction

Gilbert & Lynch used an **impossibility argument**:

- 1 Assume a system guarantees C, A, and P simultaneously.
- 2 Construct a scenario with a network partition: Node 1 and Node 2 cannot communicate.
- 3 A write w is sent to Node 1. A read r is sent to Node 2.
- 4 **Availability** requires Node 2 to respond to r .
- 5 **Consistency** requires Node 2's response to reflect w .
- 6 But **Partition tolerance** means Node 2 cannot receive w from Node 1.
- 7 **Contradiction** $\Rightarrow$ C, A, P cannot all hold



Design Implication

Since partitions **cannot be prevented** in real networks, every distributed system *must* tolerate P. The only real design choice is: **C or A** when a partition occurs.

CAP Theorem: Real System Choices & Nuances

Choice	Systems	Trade-off
CP	HBase, ZooKeeper, Mon- goDB (default)	Returns error instead of stale data during parti- tion
AP	Cassandra, DynamoDB, CouchDB	Returns possibly stale data; recon- ciles later
CA	Traditional RDBMS (MySQL, PostgreSQL)	Only works with- out network partitions (single node)

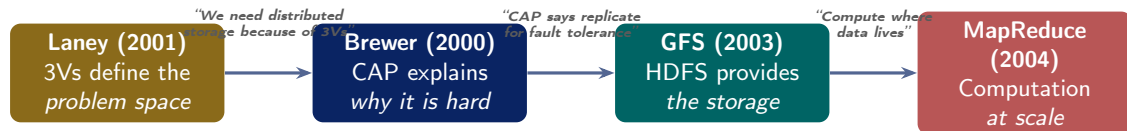
Important Nuance: Kleppmann (2015)

“Please stop calling databases CP or AP” — Martin Kleppmann, *A Critique of the CAP Theorem*, arXiv:1509.05393.

- CAP’s definitions are more nuanced than binary classifications suggest
- Real systems provide **tunable consistency** (e.g., Cassandra’s consistency levels: ONE, QUORUM, ALL)
- **PACELC model** (Abadi, 2012) extends CAP: even without partitions, there is a Latency vs. Consistency trade-off
- CAP is a **useful mental model**, not a precise engineering specification

Bangladesh Example: bKash vs. Facebook

Connecting the Papers: From 3Vs to Distributed Design



The Intellectual Lineage of This Course

Laney's 3Vs **motivated** distributed storage (because a single server cannot handle 3Vs simultaneously) → Brewer's CAP **constrained** the design space (you must choose C or A) → Google's GFS and MapReduce **implemented** a concrete CP-leaning architecture → Hadoop **open-sourced** those ideas. This entire arc from 2001–2006 forms the foundation of everything you will study in Weeks 2–10.

Case Study — McKinsey Global Institute (2011)

About the Report

Full Citation

Manyika, J., Chui, M., Brown, B., Bughin, J., Dobbs, R., Roxburgh, C., & Byers, A.H. (2011, May). **Big Data: The Next Frontier for Innovation, Competition, and Productivity**. McKinsey Global Institute.

Length: 156 pages

Downloaded: >2 million times

Cited by: >15,000 academic papers

Lead author: James Manyika, Director, McKinsey Global Institute

Scope

Surveyed Big Data potential across **five domains**: US healthcare, European public administration, US retail,

Why It Mattered in 2011

This was the first large-scale, cross-industry **quantified economic analysis** of Big Data's potential value. Before this report, Big Data was primarily a *technical* term used by engineers. McKinsey made it a *boardroom* term.

Historical Context (2011)

- Hadoop 1.0 had just been released (2011)
- Spark did not yet exist (2012)
- “Data Scientist” was named best job in America by HBR (2012)
- AWS was 5 years old; cloud computing was nascent

Key Quantitative Findings

Domain	Estimated	The Headline Argument
US Healthcare	\$300 billion+	<i>"Big data will become a key basis of competition, underpinning new waves of productivity growth, innovation, and consumer surplus. Leaders in every sector will have to grapple with the implications of big data, not just a few data-oriented managers."</i> — Manyika et al., 2011, p. 1
European Public Sector	€100 billion+	
US Retail (operating margin)	60%+ inc	
Global Manufacturing	\$600 bil	
Personal location data (EU)	\$100 bil	
US talent shortfall (deep analytics)	140,000–190,000 people by	The Talent Gap Was Under-estimated The report predicted 140,000–190,000 deep analytics jobs needed by 2018. LinkedIn's 2018 Workforce Report found 250,000+ unfilled
US manager shortfall (data-savvy)	1.5 million by	

Healthcare: The Flagship Case

US Healthcare: \$300B+ Annual Value Potential

McKinsey identified five mechanism categories:

- ➊ **Clinical decision support:** Mining EHR data to identify optimal treatments and reduce diagnostic errors. Potential: \$165 billion.
- ➋ **R&D optimisation:** Mining clinical trial data to reduce drug development cost and time. Potential: \$70 billion.
- ➌ **Public health:** Epidemic detection via syndromic surveillance of pharmacy sales and emergency room data. Potential: \$30 billion.
- ➍ **Fraud and waste detection:** Identifying anomalous billing patterns in Medicare/Medicaid claims. Potential: \$17 billion.

2024 Reality Check: What Was Achieved?

Largely realised:

- Google DeepMind's AlphaFold (2020) — solved protein structure prediction using big data + deep learning
- CMS fraud detection using ML saves \$4B/year (2022 GAO report)
- COVID-19 genomic surveillance systems (GISAID) directly applied big data to track variants in real time

Underachieved:

- EHR interoperability remains poor (data silos still exist)
- **HIPAA compliance** slows data sharing;

Critical Analysis: What Did McKinsey Get Right and Wrong?

What McKinsey Got Right ✓

- **Talent gap:** understated, not overstated
- **Competitive differentiation:** tech companies that used big data (Netflix, Amazon, Google) dominated their industries
- **Healthcare value:** clinical AI has validated many of the predicted gains
- **Retail personalisation:** Amazon's recommendation engine drives $\approx 35\%$ of revenue—enabled by big data
- **Urgency:** The hiring and investment wave that followed this report built the infrastructure now used for AI

What McKinsey Missed ✕

- **Privacy & regulation:** No mention of GDPR (2018), CCPA, or the political cost of mass data collection (Cambridge Analytica, 2018)
- **Algorithmic bias:** Big data models can *amplify* discrimination (race, gender, credit scoring)
- **Google Flu Trends cautionary tale (2008–2015):** Hyped as proof of big data power; failed catastrophically due to **overfitting and Goodhart's Law**
- **Cloud dominance:** Assumed on-premise Hadoop clusters; actual market moved to AWS/GCP/Azure

Cautionary Interlude: Google Flu Trends (2008–2015)

The Claim (Ginsberg et al., 2009, *Nature*)

“We can accurately estimate influenza activity in each region of the United States, up to two weeks ahead of traditional CDC reports, simply by tracking the volume of flu-related Google search queries.”

Initial results: Predicted flu activity with $r^2 = 0.97$ correlation to CDC data. Widely cited as proof that big data could replace traditional surveillance.

What Went Wrong

- **Overfitting:** 45 query terms selected post-hoc from 50 million possibilities
- **Goodhart’s Law:** When a measure becomes a target, it ceases to be a good measure. Media

Lessons for Big Data Practitioners

- 1 **Correlation $\neq$ causation:** A high r^2 on training data does not guarantee real-world validity
- 2 **Feedback loops:** Human behaviour changes when people know they are being measured
- 3 **Ground truth matters:** Big data **supplements**, it does not replace, rigorous data collection
- 4 **Reproducibility:** Selecting 45 features from 50M candidates without cross-validation is a **data dredging** error

Reference: Lazer et al. (2014). *The Parable of Google Flu: Traps in Big Data Analysis*.

McKinsey's Emerging-Economy Blind Spot

The 2011 report focused exclusively on the USA and Europe. But the McKinsey logic applies directly to Bangladesh:

Sector	Big Data Opportunity
Mobile finance	bKash: 65M+ accounts, real-time fraud, credit scoring for the unbanked
Garment (RMG)	Supply-chain optimisation, energy prediction in 4,000+ factories
Agriculture	Crop yield prediction from satellite data + weather APIs

A2i & ICT Division Initiatives

Bangladesh's **Aspire to Innovate (a2i)** programme is actively building national data infrastructure:

- **National Data Warehouse:** Integrating 49 ministries' databases
- **NID biometric database:** 110 million records used for KYC in banking and telecoms
- **Health Bulletin DHIS2:** District-level health data aggregation
- **Bangladesh Open Data Portal:** data.gov.bd (launched 2021)

Each of these is a real-world manifestation of the McKinsey public sector big data thesis

Guided Discussion & Live Exercise

Class Discussion Questions

Instructions

Discuss in pairs (5 minutes), then share with class. There are no single correct answers—justify your position with evidence from today's papers.

- ❶ Laney wrote his 3Vs memo in 2001 to describe e-commerce challenges. Do you think the **same three dimensions** are still the most important way to characterise Big Data challenges in 2024, or has the field evolved past this framework? What would you change or add?
- ❷ The McKinsey report predicted massive economic value from Big Data, but also **missed** the privacy, bias, and regulatory implications entirely. Does this mean we should be *sceptical* of similar economic projections for AI today? What should analysts include that McKinsey did not?
- ❸ The Google Flu Trends failure is often cited as a warning against “big data hubris.” Can you identify a current example in Bangladesh where a big data system might **fail for similar reasons** (feedback loop, overfitting, missing ground truth)?

Live Exercise: Apply the Frameworks

Instructions

Individual work, 10 minutes. Submit on paper or LMS.

Exercise A: 5Vs Classification (CLO1)

For each of the following data assets belonging to a Bangladeshi e-commerce platform, rate each V on a scale of **Low** / **Medium** / **High** and **justify in one sentence**:

- 1 Daily product sales CSV exported from MySQL at midnight
- 2 Real-time GPS location of 500 delivery riders
- 3 Customer-uploaded product review photos
- 4 HL7 medical certificate required for pharmaceutical products

Exercise B: System Design (CAP)

You are the CTO of a new Bangladeshi super-app (like bKash + Shohoz + Chaldal combined). For each sub-system below, state which **CAP trade-off** (CP or AP) you would choose and why:

- 1 **Mobile wallet balance**: showing current taka balance
- 2 **Product availability**: showing whether an item is in stock
- 3 **Ride fare estimate**: showing the price

Seminal Papers

- Laney, D. (2001, Feb 6). **3D Data Management: Controlling Data Volume, Velocity and Variety**. META Group Research Note, File 949.
- Brewer, E. A. (2000). **Towards Robust Distributed Systems**. ACM PODC Keynote, Portland, OR.
- Gilbert, S. & Lynch, N. (2002). **Brewer's Conjecture and the Feasibility of Consistent, Available, Partition-Tolerant Web Services**. *ACM SIGACT News*, 33(2), 51–59.
- Kleppmann, M. (2015). **A Critique of the CAP Theorem**. *arXiv:1509.05393*. [Recommended reading]
- Abadi, D. (2012). **Consistency Tradeoffs in Modern Distributed Database System Design: CAP is Only Part of the Story**. *IEEE Computer*, 45(2), 37–42. [PACELC]

Case Study & Critique

- Manyika, J. et al. (2011). **Big Data: The Next Frontier**. McKinsey Global Institute.
- Ginsberg, J. et al. (2009). **Detecting influenza epidemics using search engine query data**. *Nature*, 457, 1012–1014.
- Lazer, D. et al. (2014). **The Parable of Google Flu: Traps in Big Data Analysis**. *Science*, 343(6176), 1203–1205.

End of Week 1

CSE 4345: Big Data Analytics

“The world is one big data problem.”

—Andrew McAfee, MIT Sloan School

Next Week: Week 2 (Part 1 & Part 2)

Challenges of Big Data: Extraction, Integration & Heterogeneous Information

Seminal Paper: Halevy, Rajaraman & Ordille (2006) · Case Study: Healthcare Interoperability (HL7/FHIR)

Reading before Week 2: [SA] Ch. 2 · [TE] Ch. 3